



Vietnam National Universities – HCMC
International University
School of Biomedical Engineering



**DEVELOPMENT OF A WEARABLE MONITORING
DEVICE FOR PAIN ASSESSMENT BASED ON
PHOTOPLETHYSMOGRAPHY,
ELECTROCARDIOGRAPHY, AND GALVANIC SKIN
RESPONSE SIGNALS**

By

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requirements for the degree of Engineer

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LIST OF ABBREVIATIONS

ECG	Electrocardiography
PPG	Photoplethysmography
GSR	Galvanic Skin Response
SCR	Skin Conductance Response
EDA	Electrodermal Activity
RESP	Respiratory Rate
SpO₂	Saturation of Peripheral Oxygen
HRV	Heart Rate Variability
MQTT	Message Queue Telemetry Transport
TLS	Transport Layer Security
LED	Light-Emitting Diode
IR	Infrared Red
IoT	Internet of Things
BLE	Bluetooth Low Energy
CBOR	Concise Binary Object Representation
ADC	Analog to Digital Converter
I²C	Inter-Integrated Circuit Communication
IIR	Infinite Impulse Response
FIR	Finite Impulse Response



ABSTRACT

Pain remains a significant challenge in both clinical and research contexts due to the difficulty of objectively detecting it. Traditional assessment methods rely heavily on patient self-reporting, which may be inaccurate or infeasible, particularly in non-communicative patients. This thesis presents the design and development of a wearable acquisition device intended to support a machine learning–based pain detection framework using electrocardiography (ECG), photoplethysmography (PPG), and galvanic skin response (GSR) signals.

The proposed device is built around the ESP32-S2 microcontroller, which enables secure wireless data transmission via the MQTT protocol with TLS encryption. Physiological signals are acquired in real time and transmitted to a remote server, where a custom Python application performs signal processing—including filtering, feature extraction, and pain prediction—along with data visualization and storage. Device control and configuration are provided through a mobile-friendly web interface.

While the device successfully acquires and transmits high-quality physiological signals, several features remain unimplemented, including ECG lead-off monitoring and real-time pain alerts. In addition, the accuracy of the pain prediction model has not yet been validated in a clinical setting. Despite these limitations, the system establishes a promising foundation for remote bio-signal monitoring and pain assessment research, with potential for future clinical integration.

Keywords: *Wearable device; Electrocardiography; Photoplethysmography; Galvanic Skin Response; Pain assessment.*



CHAPTER 1.INTRODUCTION

1. Introduction to the biomedical problems

1.1. Pain – a subjective human experience

Pain is fundamentally a subjective human experience; however, accurately detecting its onset and levels is crucial for effective clinical assessment and treatment. The International Association for the Study of Pain defines pain as "an unpleasant sensory and emotional experience associated with, or resembling that associated with, actual or potential tissue damage" [1]. Unlike measurable physiological parameters such as blood pressure or temperature, pain cannot be directly observed or quantified. The pathway of pain involves complex neurophysiological processes that transmit signals from the periphery to the brain. Historically, pain was equated with nociceptor response to tissue damage, despite evidence of more complex underlying mechanisms [2]. Acute pain is short-term, typically lasting from a few days to under six months, and serves as a warning signal for tissue damage. In contrast, chronic pain persists beyond normal healing, often involving complex mechanisms and contributing to emotional and physical effects such as anxiety, depression, and reduced mobility [3].

The accurate assessment of pain is challenging, yet critical. Clinicians must make treatment decisions—including the type, dosage, and duration of pain relief—based on their understanding of the patient's pain. Inaccurate assessment can lead to either under-treatment—resulting in unnecessary suffering—or over-treatment—resulting in increased risk of potentially harmful or ineffective treatment, especially opioid misuse [4].

1.2. Limitations of current pain assessment tools

The current and “gold standard” pain assessment relies on patient self-reporting. A person uses tools like words, number scales, or pictures with pain scales to describe how much pain they feel. While giving useful information, self-report has several problems. A person’s pain rating is subjective and can be affected by their past experiences, thoughts, and behaviors, which are inconsistent over time. This approach is limited to use for non-communicative individuals, such as infants, patients under anesthesia, or those with cognitive impairments [5].

Behavioral observation tools are used as alternatives for non-communicative populations. These tools interpret physiological and behavioral cues—such as facial expressions, body

movements, vocalizations, and consolability—to estimate pain levels. However, observer-based assessments depend on the skill and judgment of healthcare providers [5]. Both self-report and behavioral tools provide only a single-point assessment, lacking the ability to monitor pain continuously or dynamically. These issues show the need for better, more objective ways to assess pain.

2. Analysis of current solutions

2.1. Physiological signals for pain detection

Currently, several physiological parameters such as galvanic skin response (GSR), heart rate, heart rate variability (HRV), respiration rate (RESP), blood pressure, pupillary reflex dilation, electromyogram, and skin temperature are being explored as proxies for pain [6], [7]. Two most common indices used for pain assessment in medical devices are the analgesia nociception index, which is based on HRV and the surgical pleth index, which is based on PPG [7]. GSR provides valuable information in the indication of pain [6]. The complexity of pain has led researchers to explore multimodal machine learning-based systems that combine various physiological signals to classify pain states. Teichmann et al. [8] used ECG and PPG signals with a random forest classifier for pain detection and Chu et al. [9] combined ECG, PPG, and GSR, applying linear discriminant analysis, K-nearest neighbor, and support vector machines algorithms. These studies demonstrate the effectiveness of physiological signals and machine learning in identifying pain.

a. Electrocardiography (ECG)

ECG is a non-invasive technique used to record the electrical activity of the heart over time. The heart's rhythm and contraction are regulated by electrical impulses that generate small voltage changes detectable on the surface of the body. These changes are captured using electrodes placed on the skin, typically on the chest and limbs.

The simplest form of ECG is a single-lead system. While it lacks the diagnostic capability of a standard 12-lead ECG, single-lead systems can still capture the complete ECG waveform as shown in Figure 1. Single-lead ECG enables measurement of R–R intervals, which are used to compute HRV—a key parameter for detecting physiological responses to pain stimuli.

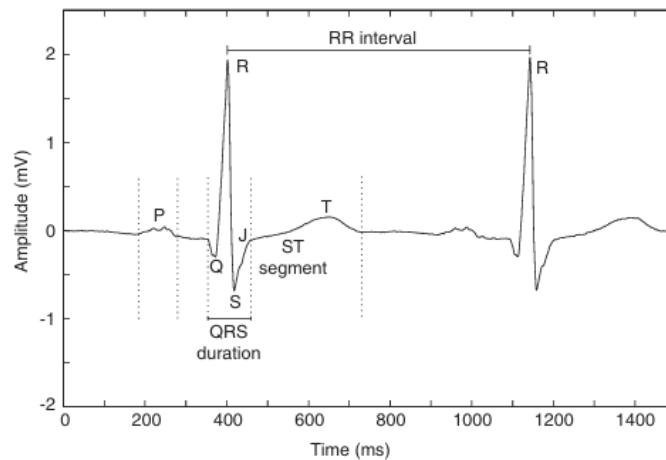


Figure 1. A normal ECG waveform, with P wave, QRS complex, and T wave [10]

b. Photoplethysmography (PPG)

PPG is a non-invasive optical technique used to detect blood volume changes in the microvascular bed of tissue. A typical PPG sensor comprises a light source and a photo detector that measures the amount of light either transmitted through or reflected by the tissue. As blood volume fluctuates with each heartbeat, the intensity of the detected light changes, producing a waveform that mirrors the cardiac cycle, as shown in Figure 2.

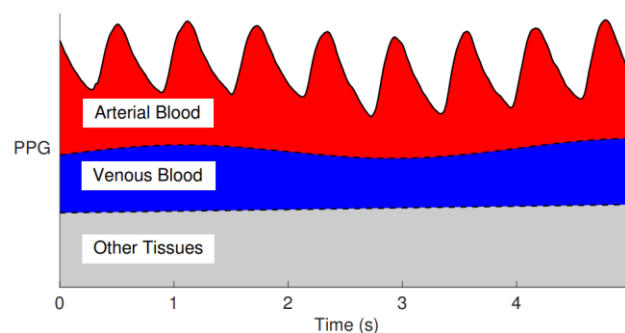


Figure 2. PPG signal and their simplified origins [11]

There are two main types of optical arrangements: transmissive and reflective. In the transmissive configuration, the LED and photodiode are positioned on opposite sides of a body part. In the reflective configuration, both the LED and photodiode are placed on the same side of the measurement site. Reflective PPG is more suitable for wearable devices and continuous monitoring due to its flexible placement, while transmissive PPG is often used in clinical-grade pulse oximeters where higher signal quality is required. PPG is typically used to estimate RESP

and oxygen saturation level (SpO_2). By analyzing the intervals between successive peaks of the PPG waveform, one can also obtain HRV and R-R intervals.

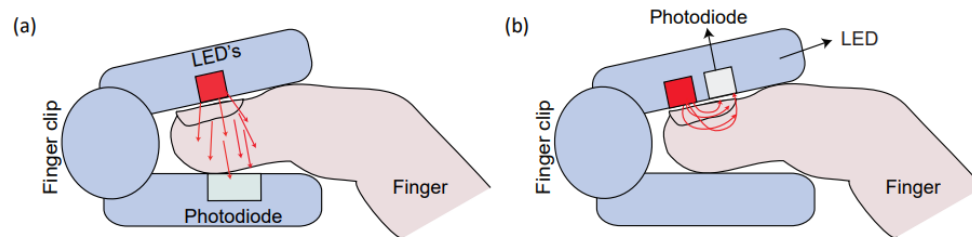


Figure 3. Transmissive mode (a) and reflectance mode (b) of contact PPG [11]

c. Galvanic skin response (GSR)

GSR, sometime called electrodermal activity (EDA), describes changes in the skin's electrical conductance caused by sweat gland activity. The tonic level of GSR is called skin conductance level; and the transient responses within seconds are skin conductance responses (SCR) [12]. GSR is widely used in psychophysiological research as a non-invasive indicator of emotional and cognitive arousal, such as stress, fear, or excitement. During emotional arousal, the sympathetic nervous system activates sweat glands, particularly in areas like the palms and soles, leading to increased skin moisture. This increases skin conductance, resulting in elevated GSR and the presence of SCR peaks.

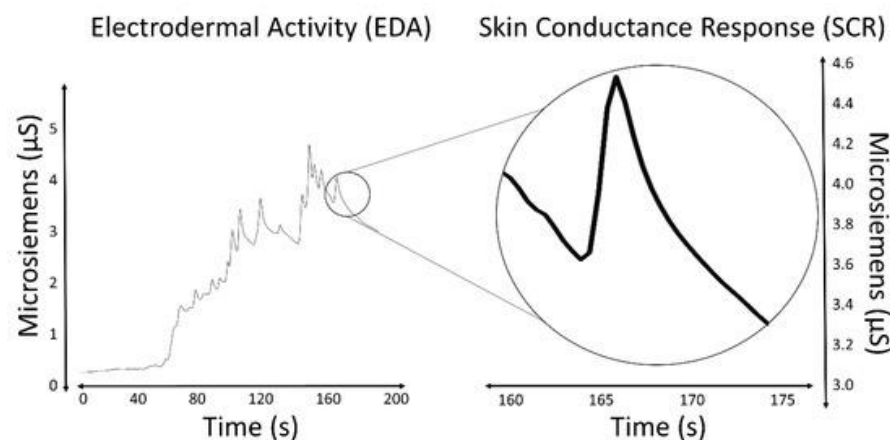


Figure 4. GSR (or EDA) signal and an isolated SCR [12]

A typical GSR setup involves placing two electrodes on the skin where sweat glands are densely concentrated, usually on the fingers or palms. Passive voltage-divider circuits are commonly used, normally with a constant voltage to measure skin conductance. A steady voltage

is applied across electrodes; the resulting current or voltage drop is used to track changes in skin conductance, as illustrated in Figure 5.

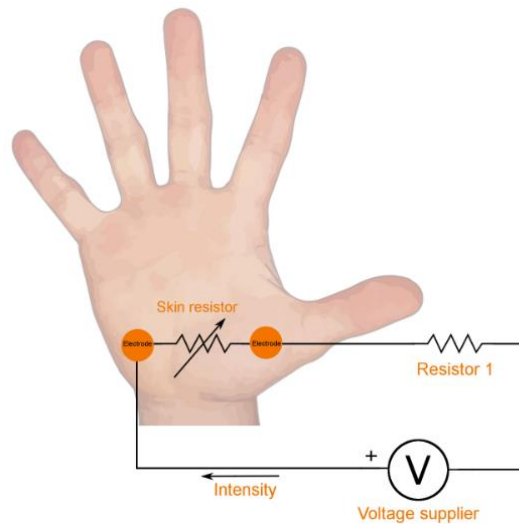


Figure 5. Simplified schematic for skin conductance measurement. Electrodes are placed on the palms or fingertips [13]

2.2. Wearable sensors used in pain detection

Table 1 and Figure 6 summarize commercialized devices and the physiological parameters for pain and stress monitoring according to Chen et al [7].

Table 1. Wearable sensors used in pain and stress detection, modified from [7]

Type of Signal	Commercialized Wearable Sensors
Heart activity	Empatica E4 wrist band, AutoSense, Cardiosport TP3, Wahoo chest belt, BioHarness 3
Brain activity	MindWave mobile EEG headset
Muscle activity	DataLOG
Electrodermal activity	Empatica E4 wrist band, BN-PPGED, Q-sensor, Shimmer sensor
Respiratory	AutoSense, SleepSense
Blood volume pulse	Empatica E4 wrist band, BN-PPGED
Body / skin temperature	Empatica E4 wrist band, AutoSense
3-axis accelerometer data	Empatica E4 wrist band, Q-sensor

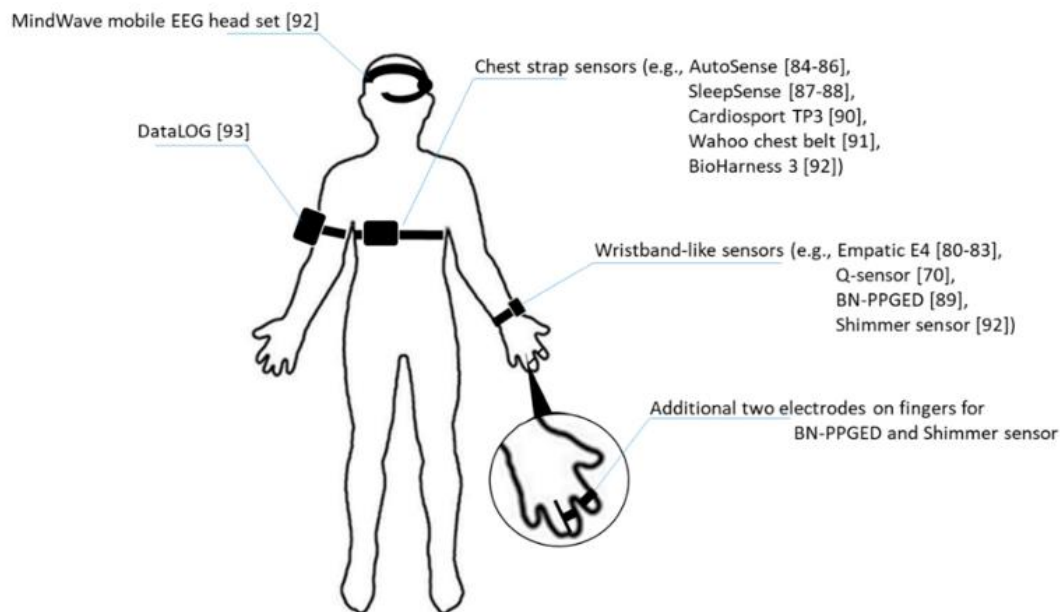


Figure 6. Illustration of pain and stress sensors placement on human body [11]

According to Table 1, the physiological signals most measured by commercial wearable devices for stress and pain research include heart activity and electrodermal activity. Blood volume pulse, measured via PPG, provides insights into cardiovascular dynamics. GSR measurement is typically performed using wristband-like sensors via two electrodes placed on the fingers to enhance signal acquisition, as seen with the BN-PPGED and Shimmer sensor.

Among eleven commercialized devices, only four include GSR measurement. Of these, only the Empatica E4 and BN-PPGED sensors also capture heart rate. The Empatica E4 stores physiological data in onboard memory for later download during follow-up visits, whereas the BN-PPGED transmits data wirelessly to a BIOPAC communication station connected to a monitoring device. While the BN-PPGED offers an ideal setup for pain assessment research, it requires a dedicated data acquisition model, making it less practical for widespread clinical use.

3. Proposed solutions

Although GSR is recognized as a valuable indicator in pain research, its limited clinical adoption has led to minimal integration into commercial devices alongside other signals, such as ECG and PPG. To address the gap, the thesis proposes a portable, easily deployable device capable of collecting ECG, PPG, and GSR signals for physiological data collection in pain assessment.

Besides the two common parameters—ECG and GSR—for pain monitoring, the PPG signal can provide a more comprehensive view into the patient physiological state by inferring SpO₂, RESP and can also be combined with ECG to estimate pulse transit time, which serves as an indirect indicator of real-time blood pressure. Beyond pain monitoring, the proposed device offers broad applications as a portable physiological data acquisition system, supporting effective health assessment and personalized care.

4. Project goals

The objective of this thesis is to design a low-cost, portable data acquisition device, referred to as *Vital-X*, capable of real-time recording of ECG, PPG, and GSR signals. The device wirelessly transmits raw data to a remote server, where signal processing, feature extraction, and pain prediction are performed. Furthermore, the device is integrated into a machine learning–based pain detection framework, demonstrating a complete and easily deployable pain assessment system suitable for clinical settings.

5. Research frameworks

The initial state of the research involves identifying existing challenges in pain management. The design requirements are then guided by relevant standards and insights from the literature review. An iterative development process enables refinement through continuous testing and evaluation. While the full system includes the training and validation of the machine learning model, this thesis focuses solely on the design of the device and its communication with the remote server for pain prediction. Figure 7 presents the research process framework that focuses on embedded system design.

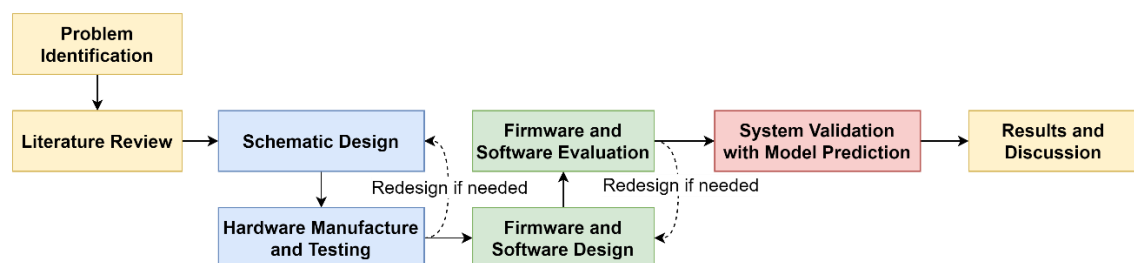


Figure 7. Research framework diagram.

CHAPTER 2. METHODOLOGY

1. System Overview

1.1. Networking protocols

Wearable devices rely on networking protocols to enable the transmission of physiological signals to external systems for processing, storage, and real-time monitoring. Figure 8 describes a typical IoT-driven system, where patients who require non-invasive monitoring due to their physiological status can be continuously monitored by the caregivers. The sensor collects the physiological information from the patient and uses gateways to further transmit the information to a cloud. This information is then sent to the caregivers wirelessly for monitoring or diagnosis.

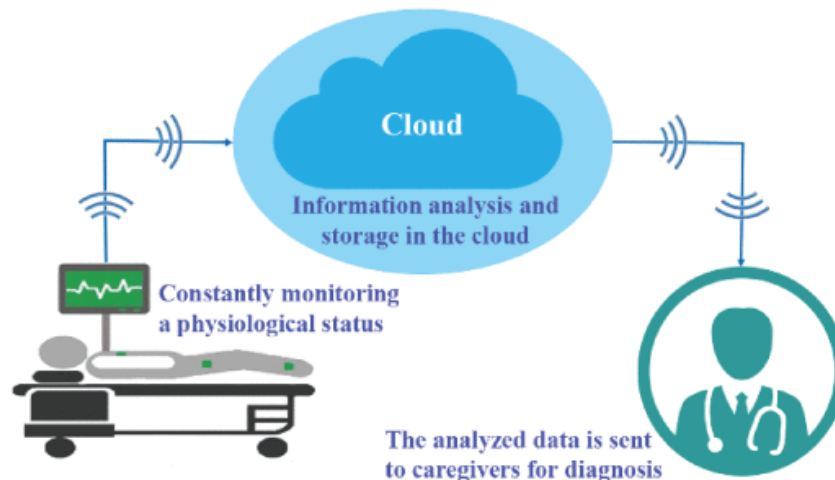


Figure 8. IoT Device in healthcare Connectivity Model [14]

Same concept was applied to the pain detection system. At the Local Area Network level, wireless protocols such as Bluetooth Low Energy (BLE), Wi-Fi, and ZigBee are commonly used to establish short-range connections between wearables and local gateway. In cloud-computing, where storage and processing happen in large data centers, internet connectivity is essential for data transmission. The simplest way to achieve this is by using Wi-Fi, eliminating the need for intermediary gateway devices required in BLE- or ZigBee-based architectures.

Table 2. Decision matrix for wireless communication protocols. Wi-Fi was selected due to its ease of integration and high data throughput.

Protocol	Max Range	Power consumption	Max Data Rate	Cloud integration	Scalability
BLE	100 m	Low (10 mW)	Moderate (1 Mbps) – suitable for transmitting small sensor data	Indirect via smartphone, which forwards data to the cloud	Moderate – suitable for small networks; limited by star topology
Zigbee	100 m	Low (35 mW)	Low (250 kbps) – sufficient for periodic sensor updates	Indirect via Zigbee hub/gateway, which connects to the cloud	High – supports large mesh networks with many nodes
Wi-Fi	200 m	High (1 W)	High (54 Mbps) - enables high-throughput applications	Direct via internet routers	Moderate – access point limits the number of concurrent devices

At the application layers - where software components communicate over the internet - common protocol stacks include Message Queue Telemetry Transport (MQTT), Hyper Text Transfer Protocol (HTTP) and Constrained Access Protocol (CoAP). Unlike HTTP, which is heavyweight, and CoAP, which can struggle with firewalls due to UDP, MQTT offers lightweight, reliable messaging over TCP, making it the good choice for efficient IoT communication [15], [16].

MQTT enables efficient communication between devices, or clients, and a central broker—which routes messages between publishers and subscribers based on topics, as shown in Figure 9. MQTT features low overhead and includes useful mechanisms such as Quality of Service (QoS), retained messages and last will messages, with strong community support from brokers such as Eclipse Mosquitto and HiveMQ.

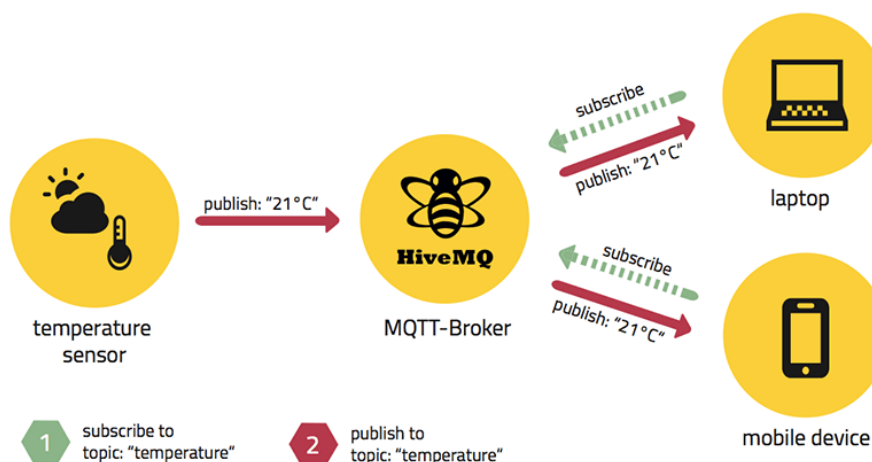


Figure 9. MQTT communication with PubSub mechanism [17]

1.2. Requirement analysis

The requirements for a medical device are typically divided into two main categories: hardware and firmware/software. Table 3 and Table 4 defines the requirement for the device firmware and hardware. The requirements follow industry standards for a Class B medical device, ensuring compliance with safety, usability, and data integrity principles (see Appendix A).

Table 3. Firmware requirements

Requirement	Description	Verification Method	Relevant Standard	Risk references
Signal acquisition	Acquire raw ECG, PPG and GSR at 512Hz	Data log review	IEC 60601-2-47	Signal distortion
Encrypted data transmission	TLS-encrypted data and transmission	Functional testing + Certificate validation	IEC 62304	Patient data leakage
Wi-Fi disconnection handling	Exponential backoff if transmission fails, maximum of 30-second delay and 5 retries. After that, the device restarts.	Disconnection test	IEC 62366 - Usability	Incomplete or corrupt logs. User confusion
LED indicator	Announce user about Wi-Fi status	Visual inspection	IEC 62366 - Usability	User confusion

Table 4. Software requirements for the mobile application

Requirement	Description	Verification Method	Relevant Standard	Risk references
User interface	Display start and stop buttons to control data acquisition Show prediction result	Visual inspection	IEC 62366 - Usability	User confusion
Collect demographic data	Send user's age, weight, height, sex to the server	Visual inspection	IEC 60601-2-25:201.12.1.103	User confusion

Table 3 outlines the embedded firmware requirements for accurate signal acquisition and reliable transmission. Since the system's pain prediction model runs on a clinical data sampled at a high frequency 512 Hz — the device's sampling rate must match this requirement to ensure data compatibility and maintain model inference accuracy.

Table 4 presents the requirements for the mobile application, or MQTT client application, responsible for device control and data visualization. Through this application, observers—such as nurses, doctors, or administrators—can enter patient demographic information, control the data acquisition process (start/stop), and observe the predicted pain level results.

Figure 10 illustrates the complete pain detection system following the successful implementation of the Vital-X device. The device publishes raw ECG, PPG and GSR data to an MQTT broker. An MQTT server subscribes to the broker to receive the transmitted data, which is then processed for pain level prediction and display of results on the observer's mobile phone.

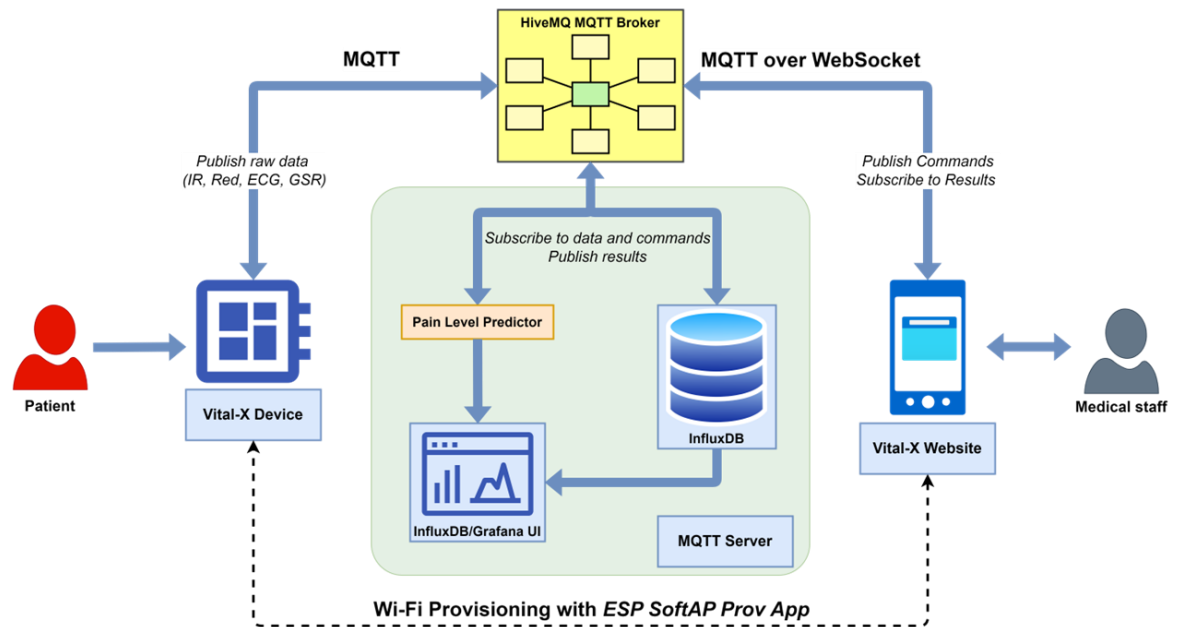


Figure 10. Conceptual architecture for implementing the Vital-X device for pain detection

2. Hardware design

Figure 11 presents a high-level overview of the hardware architecture, where the microcontroller receives analog ECG signals from the AD8232, GSR signals from a custom sensor interface, and digital PPG signals from the ADPD144RI.

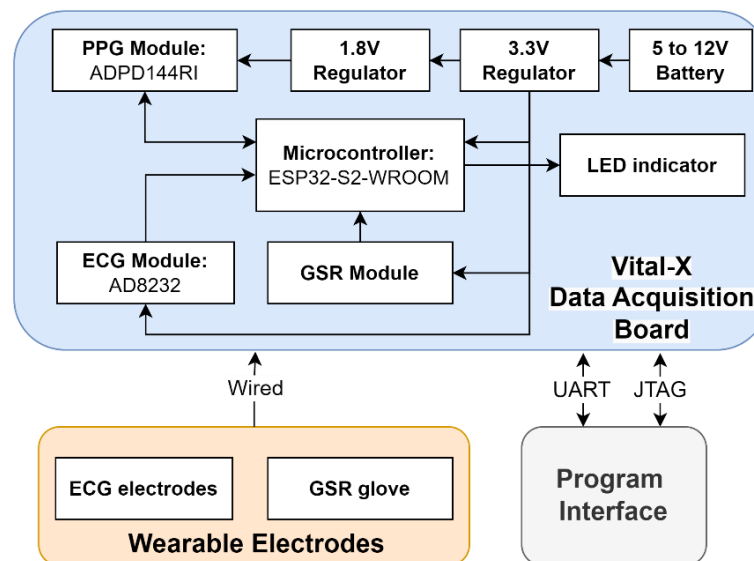


Figure 11. Block diagram of the proposed Vital-X device.

ESP32-S2-WROOM is a Wi-Fi microcontroller selected as a main control unit. It features a 32-bit single-core Tensilica LX7 CPU, integrates 2.4 GHz Wi-Fi, 4 MB of SPI flash, and has hardware-accelerated security features. An LED indicator provides visual feedback for Wi-Fi connection status. The device is battery-powered for portability.

The optional use of ESP-PROG program interface via USB-to-UART and JTAG communication allows serial programming, and advanced in-system debugging capabilities. The ECG connector interfaces with three Ag/AgCl electrodes to acquire biopotential signals from the subject. These electrodes, labeled LA (Left Arm), RA (Right Arm), and RL (Right Leg), are placed on the corresponding limbs. The GSR glove consists of two metal electrode plates positioned on the subject's fingers. A strap can be used to wear the device on the wristband. Figure 12 shows the complete and portable device setup, with battery and wearable electrodes connected to the Vital-X data acquisition board.

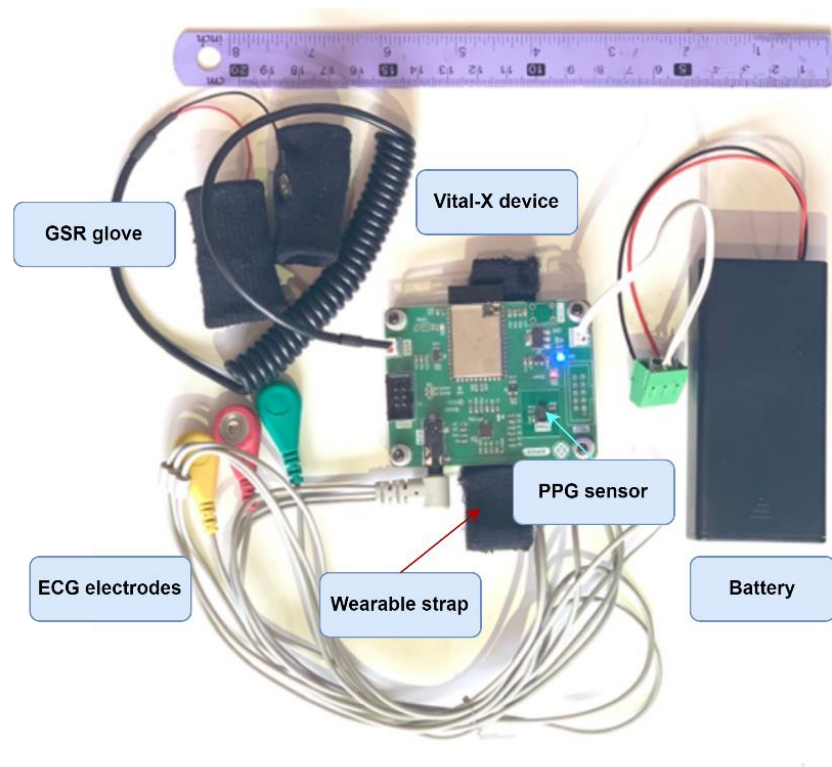


Figure 12. Complete Vital-X device setup, including housing, battery, and wearable electrodes connected for physiological signal acquisition.

3. Firmware design

3.1. Task scheduling

FreeRTOS is an open-source, lightweight real-time operation system designed for microcontrollers and small embedded systems. As shown in Figure 13, FreeRTOS is used to manage two critical tasks: sensor data acquisition and MQTT-based data transmission. A FreeRTOS software timer periodically triggers data reading from GSR, ECG and PPG sensors every $1/512$ seconds. Once the number of samples reaches the predefined `SAMPLE_BATCH` (i.e., 1536) after 3 seconds, data collection stops, and the system signals the MQTT transmission. The MQTT publish task retrieves the buffered data, initializes a CBOR encoder, encodes the sample arrays into a CBOR map, and sends the compressed payload to the MQTT broker.

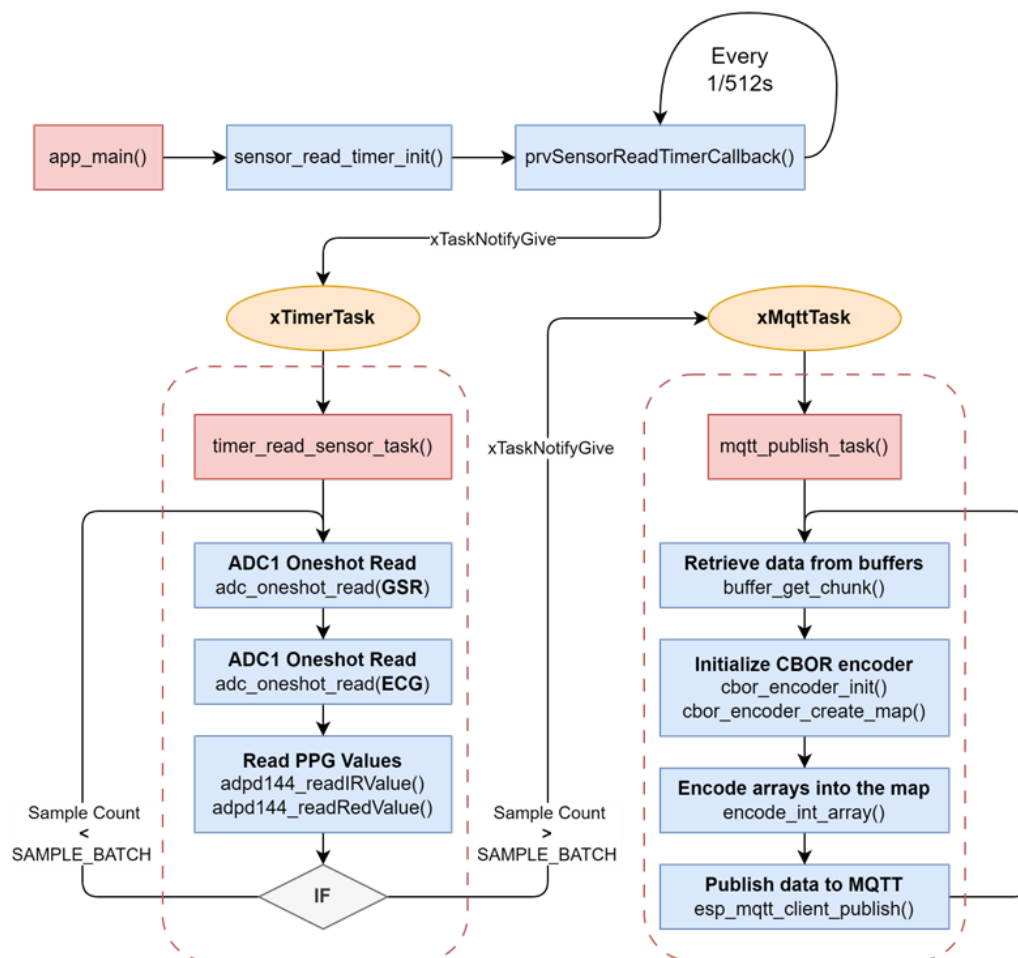


Figure 13. FreeRTOS-based firmware flowchart for data acquisition and publishing

3.2. Data acquisition strategy

The firmware for this project is developed using the Espressif IoT Development Framework (ESP-IDF), the official development platform for the ESP32. ESP-IDF provides direct access to low-level features and native support for official examples, making it particularly well-suited for complex firmware design.

a. Analog-to-Digital Converter (ADC)

The ESP32-S2 features two ADC units, ADC1 and ADC2, which are configured to acquire signals from the ECG and GSR interfaces respectively.

To maintain precise timing and alignment across all channels, a software timer triggers one-shot ADC conversions at a fixed sampling rate of 512 Hz. To ensure accuracy, a maximum 12-bit resolution was chosen. The software calibration is applied to both ADCs to produce more reliable readings. An 11 dB attenuation setting is configured for both ADC units, allowing each ADC to support an input voltage range of up to 3300 mV.

b. Inter-Integrated Circuit Communication (I²C)

The ESP32-S2 microcontroller supports two I²C interfaces, one of which is configured as the master operating in Fast Mode (400 kbit/s) and using a 10-bit addressing scheme for communication with the ADPD144RI sensor.

The sensor is configured to operate at a sampling rate of 1,000 Hz. To prevent reading overwritten value from the sensor register, data hold mechanism is employed. The readings from the infrared (IR) and red (RED) channels are triggered by a software timer at 512 Hz. The initial sensor setup, including LED power and other configurations, is provided in Appendix C.

c. Data encoding

Concise Binary Object Representation (CBOR) is a data serialization format designed for efficient message encoding. Unlike text-based formats like JSON or XML, CBOR encodes data in a compact binary form, reducing transmission size and parsing time. It supports a wide range of data types including integers, floats, strings, arrays, maps (key-value pairs), and even tagged data for semantic interpretation. CBOR-encoded MQTT messages enable fast, lightweight, and interoperable data exchange, ideal for bandwidth- and memory-constrained environments.

d. Buffering technique

To store continuous data from three sensors, a circular buffer implementation was selected. A circular buffer is a fixed-size, First-In-First-Out data structure that uses a continuous memory block with separate read and write pointers.

Inside the firmware application, there are four circular buffers which continuously receive incoming sensor data. The circular buffer has a fixed size of 2048 bytes and operates in a wraparound manner: when the buffer is full, new samples overwrite the oldest entries without interrupting the data stream. This design enables concurrent data collection and transmission without overwriting unread data. The buffer operation, integrated with CBOR encoding for transmission, is illustrated in Figure 14.

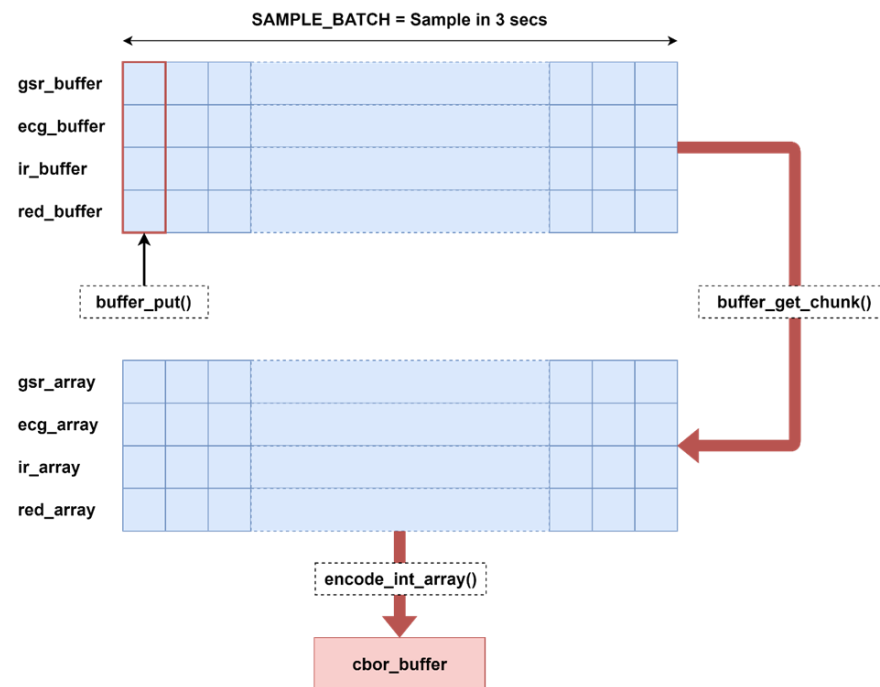


Figure 14. Circular buffer implementation to store continuous sensor data and CBOR encoding before publishing 3-second data packet over MQTT

At fixed intervals of every 3 seconds, a function named `buffer_get_chunk()` is triggered to extract a snapshot of the most recent data points stored in the circular buffer. The extracted data, consisting of integer sensor readings, is then passed to the encoding function, which serializes the dataset into a compact CBOR format, awaiting transmission.

3.3. Data transmission

a. Wi-Fi connectivity

The Vital-X device features an automated Wi-Fi connection and recovery mechanism to ensure reliable network access and MQTT communication. On power-up, it attempts to reconnect to the last known Wi-Fi network using credentials stored in the ESP32's Non-Volatile Storage (NVS). If unsuccessful, it enters provisioning mode, indicated by the onboard LED blinking at 500 ms intervals. A LED indicator serves as the main guide for users to understand the device's connection status. If the connection is later lost, the device retries connection while the LED blinks at 200 ms intervals. After five failed attempts, it clears the NVS and re-enters provisioning mode for reconfiguration. A solid LED indicates a successful connection.

To differentiate devices running the same firmware, each unit was assigned a unique identifier derived from the last three bytes of its Wi-Fi MAC address, prefixed with 'VX'. This device ID also serves as the device's Wi-Fi network name. ESP32 Wi-Fi provisioning is performed with Security Level 2 using X25519 key exchange, AES-CTR encryption, and HMAC-SHA256 authentication. This provides protection against Man-in-the-Middle attacks and eavesdropping, common risks when connecting to unsecured Wi-Fi networks. Provisioning is done via soft access point (SoftAP) Wi-Fi mode of the ESP32 using *ESP SoftAP Provisioning* application. This app is used to send Wi-Fi Network credentials, including network name and password to the ESP32 devices.

b. MQTT messaging

The ESP32 device subscribes to the MQTT command topic to receive commands from the broker. These commands control the start or stop of sensor data acquisition remotely. The device publishes responses and sensor readings to device status and data topics using QoS 0 for minimal latency. To monitor the device's availability, the firmware uses the Last Will and Testament to automatically update the device's status (e.g., "offline") after 1 minute of device's disconnection. Retained Messages retain the last known status for the patient's information and device parameters setting.

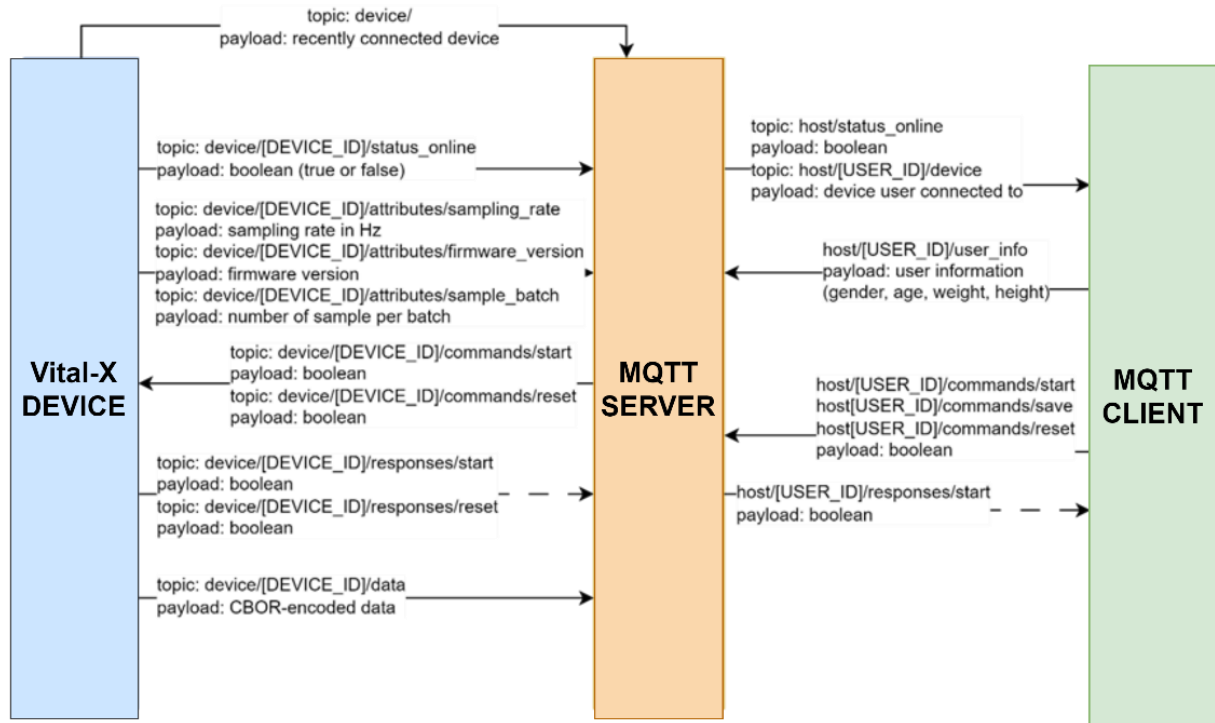


Figure 15. MQTT Topic Map for Device-Host-User Interaction. The arrow indicates topic subscribed from or published to other clients.

4. Software design

A Python application was developed on the server side. Figure 16 illustrates the data flow within the server, showing how raw data (ECG, IR, RED, and GSR) from the device is processed for pain prediction. After decoding and checking for packet loss, the raw data underwent mathematical conversion. For GSR, the voltage values (in millivolts) were converted to skin conductance (in microsiemens, μS) using a theoretical formula, shown in Appendix B.

Since the PPG sensor operates in a reflectance configuration, the captured signals—comprising both IR and RED channels—represented an inverted view of the PPG waveform. The combined PPG signal was computed using the formula:

$$PPG_{avg} = -\frac{IR + RED}{2}$$

The processed signals were then passed into the pain prediction module for feature extraction and classification every 5 seconds. For GSR, the extracted features included: amplitude,

maximum value, skewness, kurtosis, standard deviation, range, interquartile range (IQR), standard deviation of mean (SDMN), and standard deviation of successive differences (SDSD). For ECG, HRV features were extracted, including: mean R-R interval (MeanNN), standard deviation of R-R intervals (SDNN), root mean square of successive differences (RMSSD), SDSD, and standard deviation of root mean square successive differences (SDRMSSD).

All feature extraction procedures were performed using the *NeuroKit2* library [19]. The resulting features were input into a five-class ensemble classifier. The classifier combined outputs from multiple models using a voting mechanism to predict the corresponding pain level. Results — including raw data, processed signals, extracted features, and predicted pain levels — were logged into *InfluxDB*, a locally hosted time-series database, for both storage and real-time monitoring. Pain level was then published to the user interface.

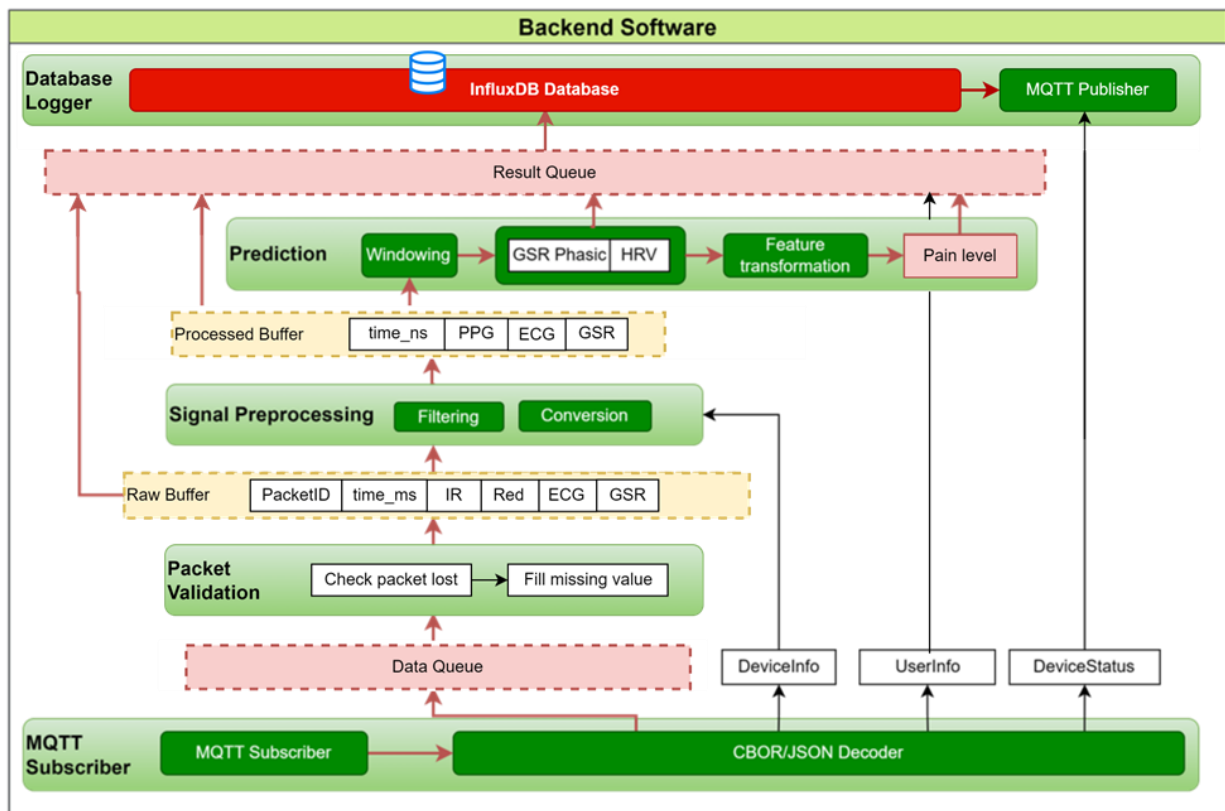


Figure 16. Software architecture, red arrows show the processing flow.

5. User Interface

The device is wirelessly configured, monitored, and controlled through a web interface hosted on *Vercel*, a cloud platform for deploying serverless web applications. Each device is accessible via a unique URL containing a query parameter `?device=DEVICE_ID`.

Figure 17 illustrates how medical staff can input patient information and specify the measurement location (Taiwan or Vietnam) through the device's web interface. This website also enables control of data acquisition, displays the device's connection status (online/offline), and provides predicted pain level.

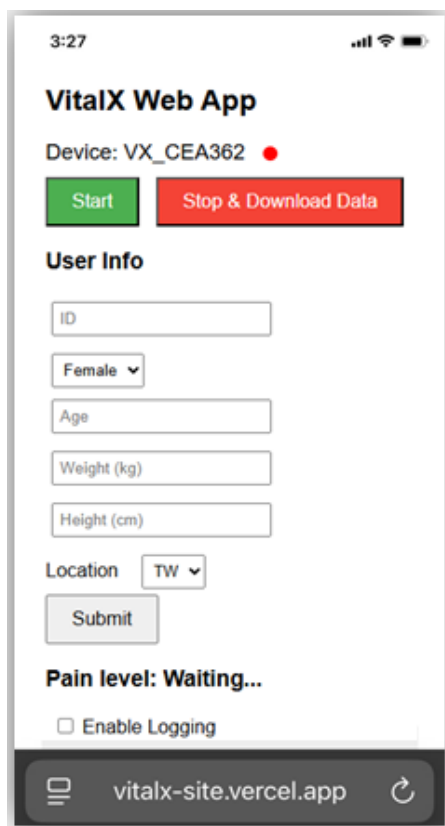


Figure 17. VitalX website for user interface.

Using ESP SoftAP Provisioning application, users can connect to Vital-X's SoftAP for Wi-Fi provisioning using either a device-specific QR code or a manual Wi-Fi connection process. Successful provisioning is indicated when the LED indicator on the Vital-X remains solid. Figure 18 illustrates the steps for observers to provision, control, and monitor the device.

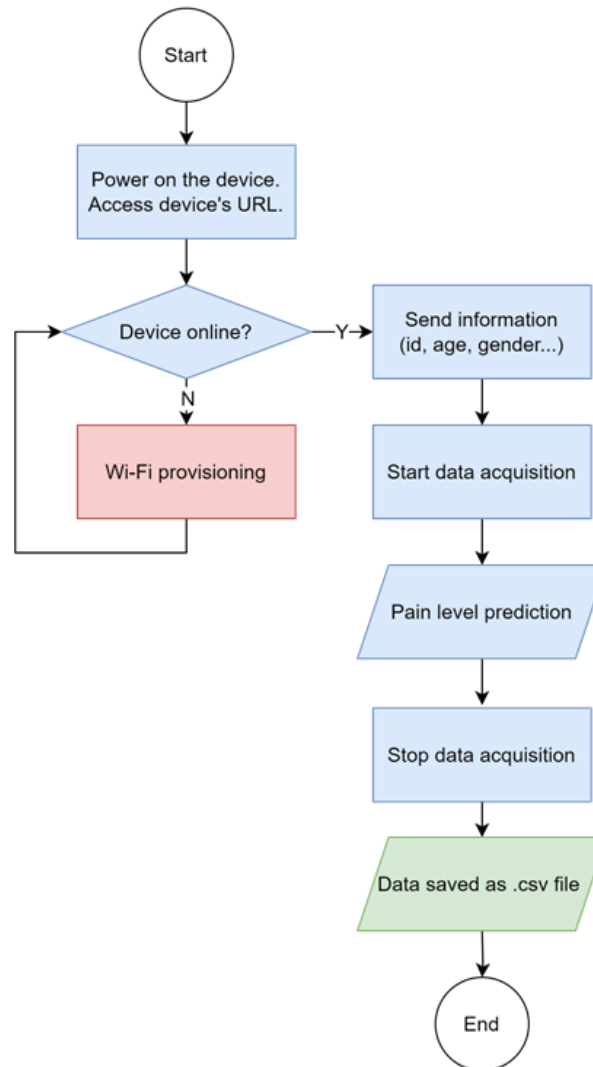


Figure 18. User guidelines for control and monitoring the device's operation

6. Test Setup and Evaluation Strategy

6.1. Equipment and Environment

The Vital-X device was tested with an ECG cable connected, using Skintact F-TB1 Ag/AgCl electrodes on the author's body. The BIOPAC MP160 system served as the reference for signal testing, and an ESP-PROG USB-to-UART interface was used for serial data logging.

A running server, a mobile phone with ESP SoftAP Provisioning and a Wi-Fi hotspot were required to test device's functionality.

6.2. Test Procedures

a. Signal Preprocessing

The ECG, GSR, and PPG signals were filtered using Neurokit's filter designs. For the ECG signal, a high-pass IIR filter at 0.5 Hz was applied to remove baseline drift, followed by a 50 Hz (for Vietnam) and 60 Hz (for Taiwan) notch FIR filter to suppress power-line interference. The PPG signal—obtained as the inverted combination of the RED and IR channels—was processed with a high-pass IIR filter at 0.5 Hz to correct for baseline wander and a low-pass FIR filter at 10 Hz to attenuate high-frequency noise such as motion artifacts. The GSR signal was filtered using a low-pass FIR filter at 3 Hz. A plot was generated to compare the raw and processed signals, demonstrating the improvements achieved through signal processing.

b. ECG Signal Evaluation

The device's ECG electrodes were connected to a human subject simultaneously with a BIOPAC system for comparative measurement. Following the American Heart Association standards for torso ECG measurement, the arm electrodes were positioned in the infraclavicular fossae medial to the deltoid insertions, while the left leg electrode was placed midway between the costal margin and the iliac crest along the left anterior axillary line [20]. The skin was cleaned and gently abraded before electrode attachment.

ECG recordings from both devices were manually time-aligned for comparability. HRV from each device was quantified using the SDNN, calculated over three successive RR intervals. Bland Altman plot was used to visualize any systematic bias and the range of agreement between the devices. This method plots the average of the two measurements against their difference,

$$Average = \frac{X_i + Y_i}{2}$$

$$Difference = \frac{X_i - Y_i}{2}$$

And the bias is the mean difference between the two measurements methods, calculated by:

$$Bias = \frac{1}{n} \sum_{i=1}^n (X_i - Y_i)$$

Where X_i is the measurement from method 1, Y_i is measurement from method 2 and n is the total number of paired observations.

The Limit of Agreement indicates the expected range for 95% of the differences between the two methods, with

$$\text{Upper Bound (UB)} = \text{Bias} + 1.96 \times SD$$

$$\text{Lower Bound (LB)} = \text{Bias} - 1.96 \times SD$$

Where Standard deviation (SD) of differences measures the spread of variability of the differences.

$$SD = \sqrt{\frac{1}{n-1} \sum_{i=1}^n [(X_i - Y_i) - \text{Bias}]^2}$$

c. PPG Signal Evaluation

For PPG, observations were made regarding the morphology of the waveform, specifically the prominence and shape of the systolic and diastolic peaks. To evaluate the consistency between cardiac features derived from ECG and PPG, R–R intervals obtained from ECG were compared with peak-to-peak intervals extracted from PPG. A correlation plot was generated to assess the relationship between the two measurements, providing a validation step for PPG-derived inter-beat interval estimation. Neurokit provides the algorithm to calculate the R-R interval. Simple peak detection algorithm was used to calculate the peak-to-peak interval from the PPG.

d. GSR Signal Evaluation

GSR measurement was evaluated by connecting precision resistors (e.g. 100 kΩ, 470 kΩ, 1 MΩ and 10MΩ) across the input terminals. For each resistor, the output was recorded over 10 seconds and averaged to obtain the mean value. The measured GSR values were plotted against the known resistors, and agreement between the measured data points and the theoretical curve was assessed.

e. Performance Test

To validate the device stability and assess signal drift, a fixed resistor of 1 MΩ was connected across the GSR input. GSR values were recorded every minute over a period of 8 hours, under moderate Wi-Fi conditions, from –40 to –50 dBm. A scatter plot was generated to identify any trends or deviations indicating drift in the signal over time. Additional functional tests were conducted in accordance with the requirements outlined in Table 3 and Table 4.

CHAPTER 3. Results

1. Signal Processing

Raw signals acquired from Vital-X device include infrared (IR) and red (RED) channels, as shown in Figure 19(a) and Figure 19(b). These waveforms reflect pulse wave activity but do not resemble a standard PPG waveform due to the sensor's reflective configuration. Figure 19(c) shows the raw PPG signal obtained by taking the inverted average of the IR and RED signals. A filter was then applied to the raw PPG signal to remove the slowly declining DC component, enhancing the visibility of the pulsatile component associated with cardiac activity.

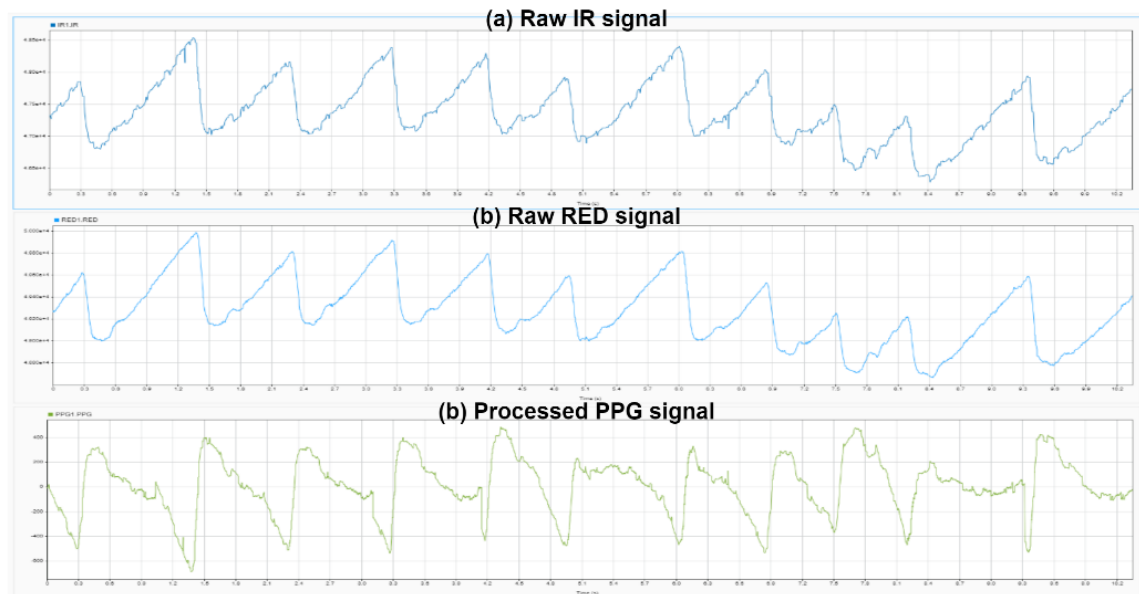


Figure 19. PPG signal processing: IR channel, red channel and derived PPG signals

Figure 20 and Figure 21 shows a clear improvement in signal quality after filtering. Figure 20 presents the raw ECG output from the AD8232 sensor which is heavily contaminated by high-frequency noise, shows the improved ECG signal after applying digital notch filters at 50 Hz and 100 Hz, targeting power-line noise and its harmonic.

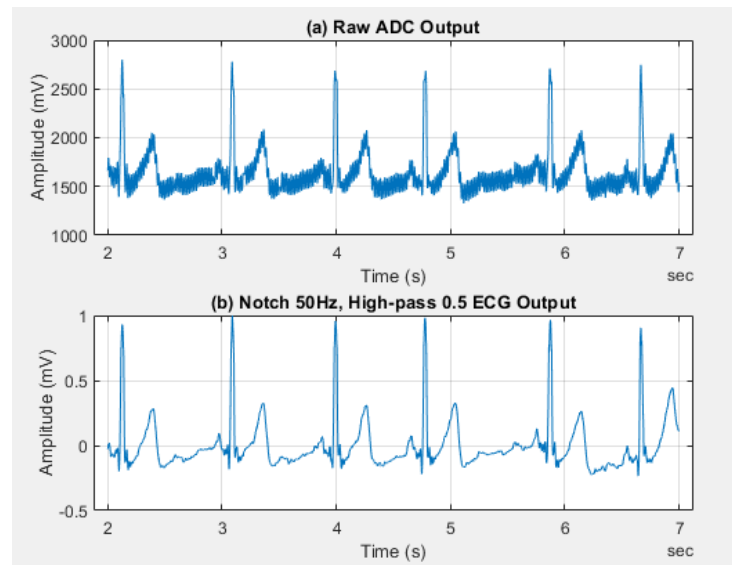


Figure 20. ECG signal processing: Raw and filtered.

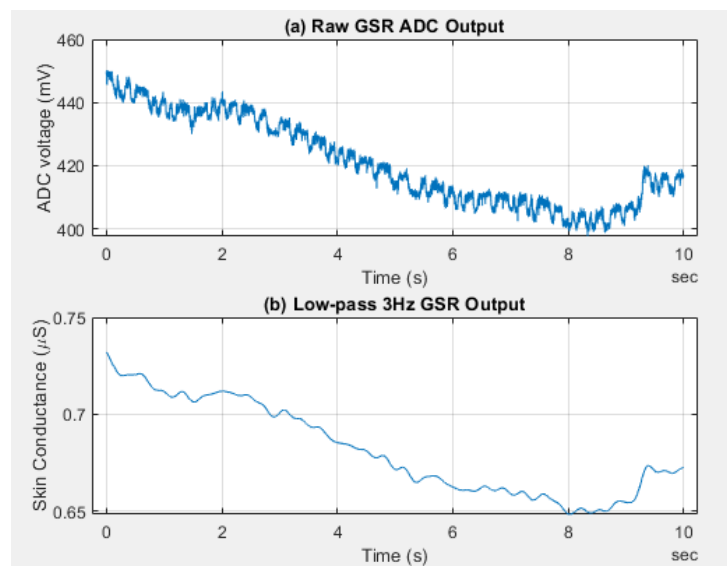


Figure 21. GSR signal processing: Raw and filtered

2. ECG Signal Evaluation

Figure 22 shows the ECG waveforms recorded from BIOPAC and Vital-X on the same plot, showing similar waveform shapes between the two devices. The Bland–Altman plot indicates strong agreement between the devices' algorithm, with a small bias of 11 μs and a narrow upper and lower limits of agreement at 127 μs and $-105 \mu\text{s}$, respectively.

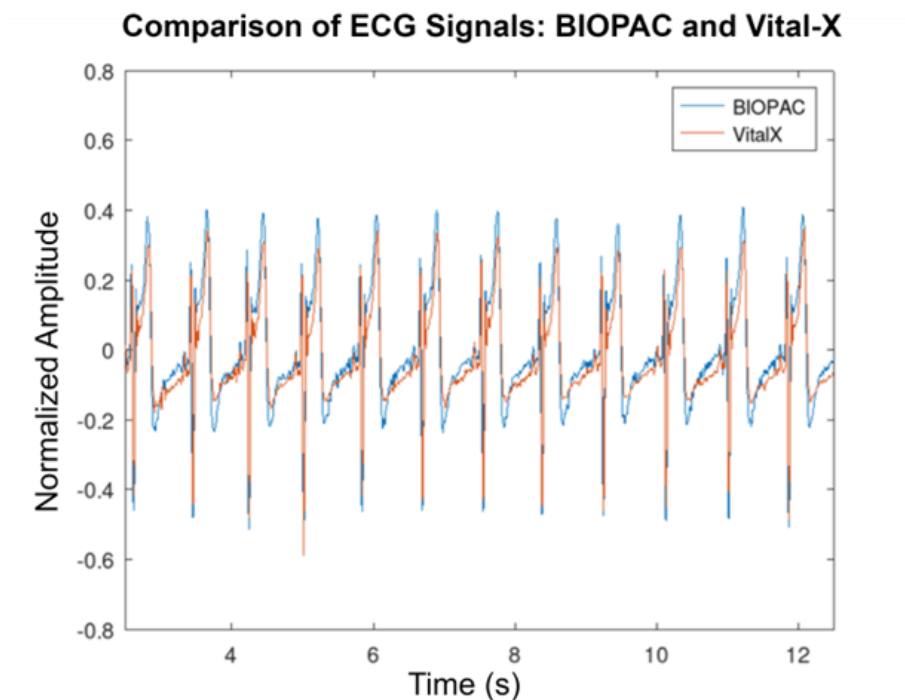


Figure 22. Overlay plot of ECG acquisition by Vital-X and BIOPAC.

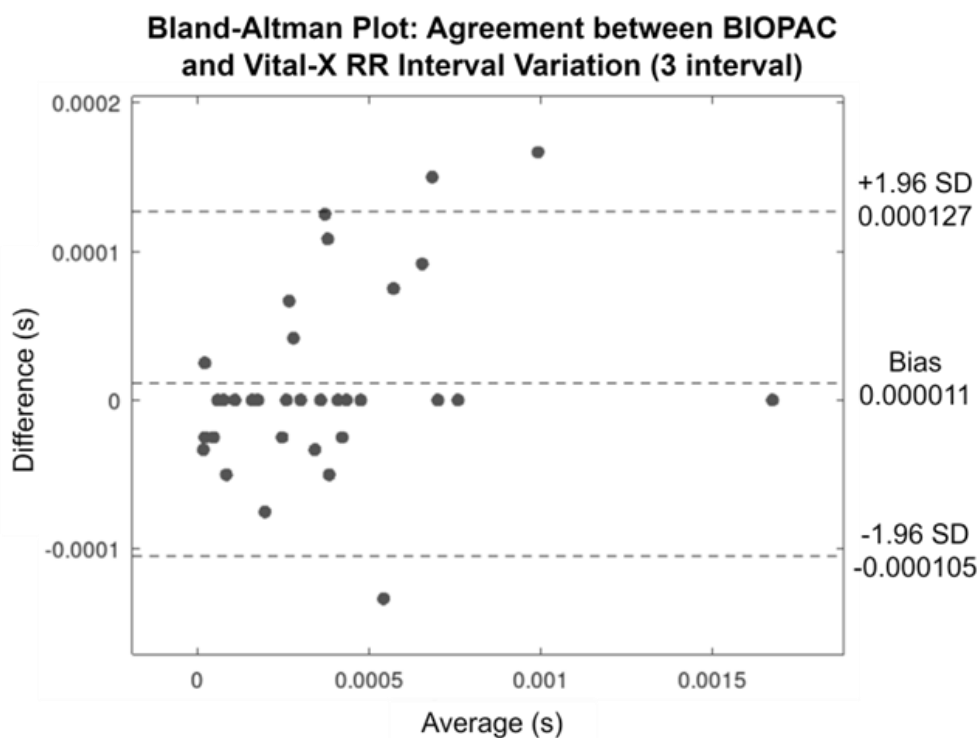


Figure 23. Bland-Altman plot comparing HRV measurement from Vital-X and BIOPAC

3. PPG Signal Evaluation

Figure 24 compares PPG signal and ECG signal recorded simultaneously. The PPG waveform shows key features like the systolic peak and diastolic notch. t_{RR} denotes the R-R interval, while t_{PP} represents the interval between two consecutive systolic peaks in PPG. A close match between t_{RR} and t_{PP} values was observed. Figure 25 shows a strong correlation ($R = 0.955$) between ECG R-R intervals and PPG peak-to-peak intervals, with a mean absolute percentage error (MAPE) of 3.19%.

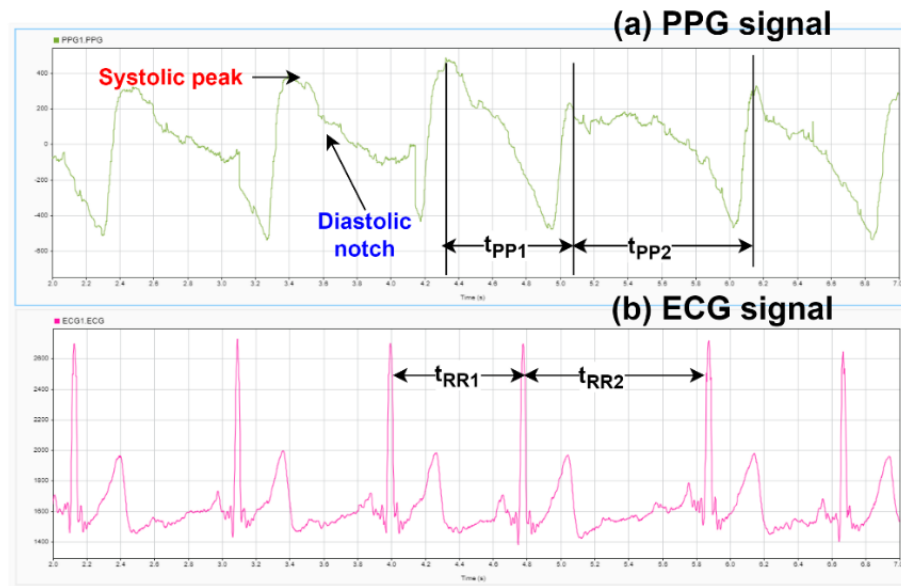


Figure 24. PPG and ECG signals acquired simultaneously using Vital-X

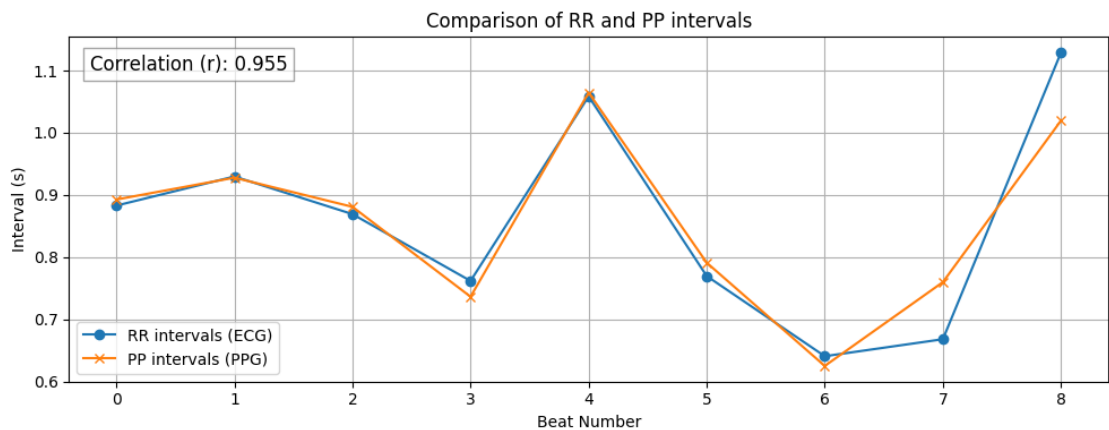


Figure 25. Correlation between ECG R-R intervals and PPG P-P intervals, by Vital-X.

4. GSR Signal Evaluation

Figure 26 illustrates the relationship between input resistance and GSR voltage. A constant reading at resistance values less than 100 k Ω suggesting ADC saturation at 2583 mV. The mean absolute percentage error of all measurements is 1.25%, indicating high accuracy between theory and measurement.

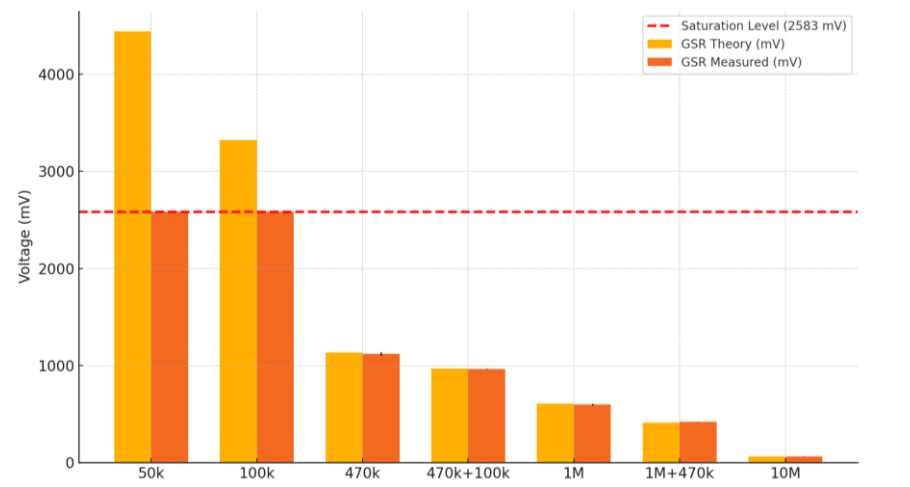
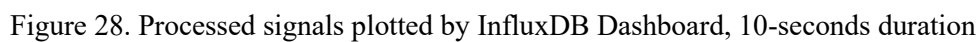
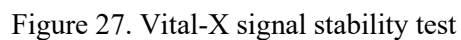


Figure 26. GSR signal measurements compared to theory

5. Performance Test

Figure 27 presents the variation of GSR measurements recorded using the Vital-X device over a period of 8 hours. Each blue circle represents an individual GSR reading plotted every minute. The red line indicates the mean GSR value. The time-series plot of the signal showed no observable trend, with no packet loss. The GSR readings demonstrated high stability over an extended period, with a mean deviation of just 0.14 mV.

The Vital-X device is capable of remotely communicating with a server, continuously transmitting data packets of 3-second intervals in real time. Figure 28 presents the InfluxDB dashboard on the server side, which displays real-time bio-signal visualization and pain prediction results.



CHAPTER 4. DISCUSSION

The Vital-X device demonstrates several strengths as a low-cost, real-time physiological monitoring system for pain detection. A key advantage lies in its ability to reliably transmit data via Wi-Fi, achieving no packet loss over an 8-hour period. This direct communication with a remote server eliminates the need for intermediate gateways, which are commonly required by commercial devices that rely on short-range Bluetooth and local data storage. By offloading computational tasks such as feature extraction and machine learning inference to the server, Vital-X maintains efficient performance despite its limited onboard processing resources.

The quality of acquired physiological signals supports the system's reliability for pain assessment. HRV extracted from ECG recordings closely aligned with results from a reference BIOPAC system, as confirmed by Bland–Altman analysis. Similarly, PPG peak intervals showed strong correlation with ECG R–R intervals, validating PPG-derived inter-beat estimation. The custom-designed GSR module yielded stable readings consistent with theoretical expectations.

In contrast to compact wristband devices such as the Empatica E4, Vital-X requires more cumbersome electrode placement for both ECG and GSR acquisition. Moreover, the absence of an onboard accelerometer makes the device more vulnerable to motion artifacts, limiting its suitability to contexts in which patients remain immobile, such as during surgical procedures or post-anesthesia care unit stays.

Another limitation is ADC saturation in the GSR measurement circuit. When using low-resistance values (e.g., 100 k Ω), the output voltage reaches the ADC's maximum of 2.853 V, resulting in signal clipping and loss of meaningful data. This threshold is relatively high for typical human skin resistance, ranging from 25 k Ω to 1 M Ω [21], especially under conditions of high sweat gland activity.

Overall, these findings indicate that Vital-X can provide reliable, high-quality physiological data streams suitable for real-time feature extraction and vital sign monitoring. By combining stable wireless transmission, efficient server-based processing, and accurate bio-signal acquisition, the system demonstrates strong potential for deployment in clinical settings where cost-effective and scalable pain monitoring solutions are required.



CHAPTER 5.CONCLUSION.

1. Summary of the research development

This thesis presents the design and implementation of Vital-X, a low-cost, portable data acquisition device aimed at real-time pain detection using physiological signals. The system captures raw ECG, PPG, and GSR signals at a sampling rate of 512 Hz and transmits the data via Wi-Fi to a remote machine learning inference server for signal processing, feature extraction, and pain prediction. Device control and Wi-Fi provisioning are managed through a mobile phone website and application, while MQTT is used as the messaging infrastructure to ensure real-time communication among the device, the server and the mobile phone.

2. Limitations

Several limitations constrain the practicality and clinical applicability of the device. Reliable operation depends on stable Wi-Fi connectivity, which in hospital environments may require dedicated access points or room-specific hotspots. The use of two separate mobile applications—for Wi-Fi provisioning and device control—further adds to the complexity of the user experience. In addition, key features remain unimplemented, including ECG lead-off detection and real-time pain alert functionality, which limits both the safety and clinical utility of the system. Finally, the accuracy of the pain prediction model has not yet been validated, underscoring the need for further evaluation and refinement.

3. Suggestion and future application

Despite existing challenges, the proposed system architecture lays the groundwork for several promising directions, including continuous pain monitoring to guide medication adjustment and optimize patient care. Future work could further enhance the system's flexibility, scalability, and clinical integration by enabling interoperability with electronic health records via HL7/FHIR standards, thereby facilitating seamless data exchange within clinical workflows. In addition, the device could be extended to infer other vital signs, such as heart rate, blood pressure, SpO₂, RESP, and body temperature from the ECG and PPG signals. These developments would strengthen the device's role as a comprehensive physiological monitoring platform and support broader adoption in diverse clinical contexts.



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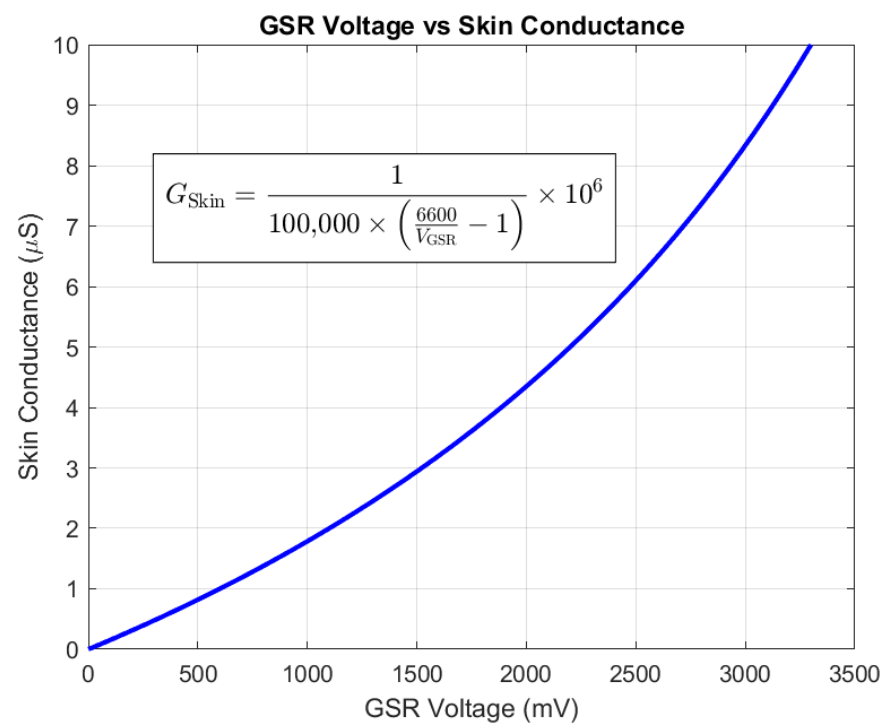
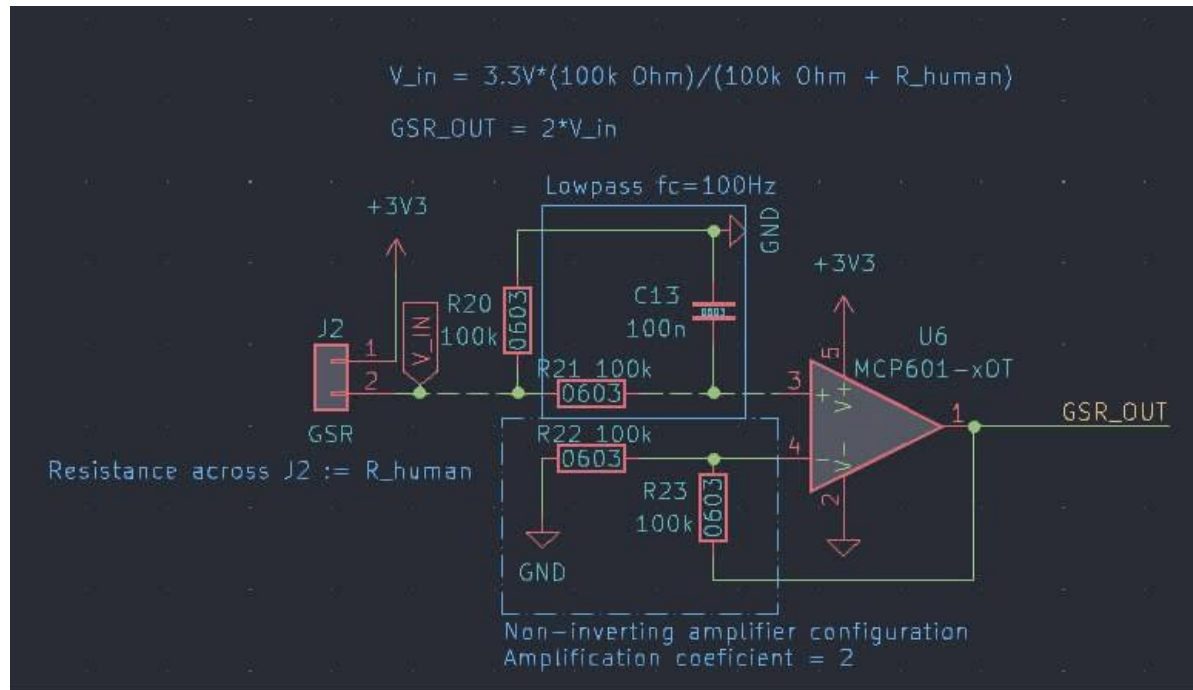
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APPENDICES

Appendix A. Requirement Analysis for a Class II Wearable Device with Class B software

Requirement	Description	Acceptance Criteria
IEC 60601-1 Safety	Electrical safety, thermal safety, mechanical safety, data safety, electromagnetic compatibility	Patient leakage current: <10 uA Enclosure leakage current <100 uA Insulation resistance > 5 MOhm Temp: < 41 – 43 C for contact area No sharp edge, has adjustable strap Encrypted data storage and transmission No failures during immunity testing
IEC 62304 - Software development	Software development, risk management, configuration management	Software safety classification Version control, development environmental setup, task management. Functional and non-functional requirements. System block diagram, Modular software architecture Programming language, Coding standards, error handling strategy Unit, integration and system testing Identify software hazards and mitigations (ISO 14971) Version identifier & Changelog Appendix for traceability matrix
IEC 62366 - Usability engineering	Human factors and safety-related usability	Intended users, use environment Potential use errors, potential harm and implement risk controls by UI or instructions
IEC 60601-2- 47 – Ambulatory ECG	ECG signal fidelity, Data recording and storage, Lead systems	Bandwidth: 0.05 Hz – 100 Hz; sampling rate $\geq$ 200 Hz ADC resolution: $\geq$ 10 bits Timing accuracy: drift < 1 second over 24 hours RR detection error: $\leq \pm 5$ ms compared to reference Data loss: 0% during normal operation Input impedance: Specifies battery performance and safety under extended operation Proper lead labeling, color-coding, and secure connection Lead disconnection detection
ISO 80601-2- 61 – Pulse oximeter	Relevant safety and performance clauses for heart rate monitoring	Pulse rate accuracy: ± 3 bpm or $\pm 3\%$, whichever is greater Sampling rate: ≥ 25 –50 Hz (raw PPG) LED power safety

Appendix B. Vital-X Schematics and GSR theoretical curve



Appendix C. ADPD144 PPG sensor initial register configuration

Register Address	Value	Register Name	Description
0x10	0x0001	Mode	Set to 0x0001 to enable writing to registers.
0x02	0x0005	INT_IO_CTL	Enables interrupt drive, active low.
0x11	0x30A9	SLOT_EN	Enables FIFO drop if full. Time Slot A & B = sample mode with 32-bit sum.
0x12	0x0001	F_SAMPLE	Sample rate = 1000 Hz.
0x14	0x0116	PD_LED_SELECT	PD1–PD4 to Channel 3 & 4; Time Slot A & B active; LED2 pulses in A, LED1 in B.
0x15	0x0330	NUM_AVG	Average 8 samples in Time Slot A and B.
0x18	0x3FFF	SLOTA_CH1_OFFSET	Not used (max value 16383).
0x19	0x3FFF	SLOTA_CH2_OFFSET	Not used (max value 16383).
0x1A	0x1FF0	SLOTA_CH3_OFFSET	Offset = 8176 (half-scale).
0x1B	0x1FF0	SLOTA_CH4_OFFSET	Offset = 8176 (half-scale).
0x1E	0x3FFF	SLOTB_CH1_OFFSET	Not used (max value 16383).
0x1F	0x3FFF	SLOTB_CH2_OFFSET	Not used (max value 16383).
0x20	0x1FF0	SLOTB_CH3_OFFSET	Offset = 8176 (half-scale).
0x21	0x1FF0	SLOTB_CH4_OFFSET	Offset = 8176 (half-scale).
0x23	0x3005	ILED1_COARSE	Scale 100%, driver current = 100 mA.
0x24	0x3007	ILED2_COARSE	Scale 100%, driver current = 130 mA.
0x25	0x0207	ILED_FINE	LED1 & LED2 fine timing = 593.75 ns.
0x30	0x0319	SLOTA_LEDMODE	Pulse offset A = 25 μ s, pulse width = 3 μ s.
0x31	0x0813	SLOTA_NUMPULSES	Pulse period A = 19 μ s, pulse count = 8.
0x35	0x0319	SLOTB_LEDMODE	Pulse offset B = 25 μ s, pulse width = 3 μ s.
0x36	0x0813	SLOTB_NUMPULSES	Pulse period B = 19 μ s, pulse count = 8.
0x39	0x21F3	SLOTA_AFEMODE	AFE fine offset = 593.75 ns, AFE offset = 15 μ s, AFE width = 4 μ s.
0x3B	0x21F3	SLOTB_AFEMODE	AFE fine offset = 593.75 ns, AFE offset = 15 μ s, AFE width = 4 μ s.
0x42	0x1C36	SLOTA_GAIN	TIA = 50k, band-pass = 390kHz/100kHz, VREF = 1.265V, integrator = 200k/6.33pF.
0x44	0x1C36	SLOTB_GAIN	Same as Slot A Gain.
0x4E	0x0040	ADC_TIMING	ADC clock only pulses during conversion.
0x10	0x0002	Mode	Set to 0x0002 to run the device.